

Bitcode

\$BTD — Tokenomics

Canonical · leftover-mint · dynamic self-sustaining fees. Where this document and BITCODE_SPEC diverge, SPEC wins.

1 · What \$BTD is

\$BTD is Bitcode's knowledge-currency: a fungible, freely tradable token whose supply is tied to the industrial value of data that has been paid for.

Sellers do not set prices. Bitcode measures the data — its quality and its quantity — and reduces those measurements into a single scalar: \$BTD. One may think of \$BTD as **knowledge-backed currency**.

The most meaningful measurements for commercial value are made when buyers request the kind of knowledge they intend to buy. Those Need-relative measurements alone are summed. How much the deposited material satisfies the Need is the \$BTD of the DataPack: *the knowledge volume of that data slice under that demand*.

2 · Settlement, election, and the mint

Buyers do not pay in \$BTD. They settle with other capital — ETH, BTC, SOL, or other admitted payment rails.

After purchase settles, the seller elects how to take their payout: a division between the payment currency and \$BTD. The election is a free continuum, 0–100%, set as a standing preference and overridable per settlement. **Only the elected \$BTD share is minted.** The share of the measured scalar the seller declines is never created — not burned, not deferred, simply never in existence.

The election has a counterparty. Choosing \$BTD means forgoing the corresponding share of the cash, and that forgone cash flows to the Bitcode treasury. This is what makes every minted token *paid for*: its price in real capital is sitting behind it. On top of this counterparty flow, the treasury takes a small dynamic fee from both payout buckets (\$4).

Formally, for a settled read of one DataPack, with fee rate f :

P = settlement paid by the buyer (payment currency, e.g. 1 ETH)

B = measured knowledge volume of the pack under the Need (e.g. 1 \$BTD)

s = seller's elected payment-currency share, $0 \leq s \leq 1$

Seller receives: $(1-f) \cdot s \cdot P$ payment currency
+ $(1-f) \cdot (1-s) \cdot B$ \$BTD, minted at settlement

Minted this tx: $(1-s) \cdot B$ — the elected bucket, nothing else

Never minted: $s \cdot B$ — forgone issuance, permanent

Treasury receives: $(1-s) \cdot P$ — forgone cash: the mint's backing
+ $f \cdot s \cdot P$ — cash-bucket fee
+ $f \cdot (1-s) \cdot B$ — \$BTD-bucket fee

Worked example — cost 1 ETH, measured scalar 1 \$BTD, seller elects 90% cash, $f = 1\%$: the seller takes **0.891 ETH + 0.099 \$BTD**; total mint is **0.1 \$BTD** — the 0.9 \$BTD of measured volume the seller declined is never created; the treasury takes **0.109 ETH + 0.001 \$BTD**.

Minting after payment binds \$BTD's supply to the industrial value of data in the Depository: every token in existence corresponds to measured knowledge that real capital already paid for — and that a seller, the party best positioned to judge the data's worth, chose to hold as knowledge rather than cash.

3 · What leftover-mint means for the token

The election is an issuance governor with three consequences.

Doubly-backed supply. A pair-wise mint would back every token with capital spent. The leftover mint adds a second backing: seller conviction. \$BTD enters existence only when a seller prefers the token to money, and the money they declined lands in the treasury behind it. Issuance is a revealed-preference signal, not mechanical printing. (The fee-share of mint the treasury takes is the one exception to “every token was chosen by a seller”; at fee scale it is a rounding term on the story.)

Structurally scarcer. Aggregate issuance is $\sum (1-s_i) \cdot B_i$ across all settlements — always at most the measured volume transacted, usually well under it. Every cash-preferring seller permanently forgoes issuance. The supply approaches its cap only as fast as sellers collectively choose to hold knowledge, with no burn mechanism needed.

A conviction flywheel, not a printing press. If \$BTD appreciates, sellers elect more of it — supply grows precisely when demand to hold it is strongest, and the marginal token is still fully paid for. If sentiment weakens, elections shift to cash and issuance self-throttles. Supply growth is pro-conviction rather than pro-cyclical.

4 · The self-sustainability invariant (dynamic fees)

Running the exchange costs real resources: computing measurements (static analysis, sandboxed execution, provenance indexing, judge inference), storing artifacts and evidence roots, settling on-chain, and maintaining the versioned reference corpora and models that keep measurements honest.

The treasury's intake is required to cover them:

$$E[(1-s) \cdot P + f \cdot s \cdot P + f \cdot (1-s) \cdot B] \geq E[C]$$

**C = per-settlement cost of measurement compute, artifact storage,
settlement execution, and reference upkeep (trailing average)**

The fee f is **dynamic**: it floats (bounded, smoothed over a trailing window) at whatever level keeps this invariant true. Bitcode is therefore at all times at least self-sustaining — the market pays for its own instrument, and the measurement engine can never be starved into dishonesty.

The invariant gives the fee an elegant character: it is **countercyclical to conviction**. When sellers elect mostly \$BTD, forgone cash funds the treasury and fees rest near their floor. When sellers elect mostly cash, the forgone flow thins and fees rise modestly to carry the costs. The operating budget is guaranteed in both regimes, and in neither does it depend on token appreciation, external subsidy, or discretionary intervention.

5 · Supply schedule

Like Bitcoin, \$BTD has a finite supply of the same order of hard cap (21M-class), minted under a supply-decay as cumulative issuance approaches that cap. The decay applies to the minted amount itself:

$\text{mint} = \text{decay}(\text{issued}) \cdot (1-s) \cdot B$ **decay: smooth, $\rightarrow 0$ as issued $\rightarrow$ cap**

Every settlement mints something; the cap is an asymptote, never a cliff. And because the leftover mint already under-issues relative to measured volume, the decay binds later than it would under a pair-wise design — the cap is approached only by the cumulative choice of sellers to hold knowledge.

On-chain, Bitcode is ERC-1155: token id 0 is fungible \$BTD; ids ≥ 1 are DataPack co-ownership. Each asset keeps its own tokenomics — supply, security, markets — and both trade freely and globally. \$BTD mints only on **primary** settlement; secondary trading of packs or of \$BTD itself is pure market activity and mints nothing.

6 • Price and volume: paths under consideration

P (what the buyer pays) and B (the measured knowledge volume) are distinct quantities. How they relate is a launch decision; the three coherent paths:

Path 1 — bidding sets P ; B is volume. Price discovered by the live market; B is what mints. Preserves real price discovery — the live-exchange differentiator — and yields $P \div B$ as a publishable market statistic: an emergent **price of knowledge** index, arguably a product in itself. Its cost is drift: thin early liquidity prices noisily, and P and B can diverge enough to strain the knowledge-backed narrative. Mitigation is transparency — publish $P \div B$, monitor drift, never hard-peg.

Path 2 — P derived from B . Price quoted from the measured scalar. The tightest possible knowledge-backed claim and the simplest buyer experience, but there is no market, only an oracle — and every measurement-gaming attack becomes a pricing attack. Plausible as a launch training-wheel; wrong as an end state.

Path 3 — hybrid floor. Bidding above a B -derived floor. Discovery preserved above the floor; measurement guarantees knowledge never clears below a bound — protection for sellers in thin early markets, with a graceful dial (lower the floor coefficient as depth arrives) toward Path 1.

The natural sequencing: **launch near Path 3, converge to Path 1** as liquidity deepens — Path 2 surviving only as the floor formula inside 3, never as the market.

7 • Anticipated movements

Design positions taken now so later changes are evolutions, not reversals:

Settled amounts never restate. Reference bumps (corpus, frontier, judge versions) re-measure forward only; minted \$BTD is final.

Multi-seller packs. Each contributor elects their own s over their contribution-weighted share of B — contribution measured from the material, not self-declared.

Treasury \$BTD policy. Fee-accrued \$BTD is held under a published policy (hold, fund, or market operations) — the treasury never mints outside the fee path.

Fee governance. f 's bounds, smoothing window, and the cost basis C are SPEC constants; changing them is a SPEC-gated event, not an operational dial.

New rails. Additional payment currencies admit without touching mint mechanics — P is rail-agnostic by construction.

Deeper on-chain measurement proofs. Evidence roots and replayable disputes extend the settlement layer without changing issuance.

8 • Finalization ledger

Remaining launch constants — everything else in this document is settled design:

Open: fee bounds + smoothing window for dynamic f · cost-basis C definition · exact decay function and hard-cap figure · $P \leftrightarrow B$ path selection (§6 sequencing lean stated).

Leaning: 18 token decimals; dust floor tied to the minimum measurable fit.

Advanced Engineered Software, Inc. · Canonical from July 27, 2026 · Propagate leftover-mint + dynamic-fee language to: one-pager, appendix A8, marketing Panel 2, Cryptography slide. Measurement mechanics: measurement master doc · gaming defenses: anti-gaming doc — this document owns issuance, election, treasury, and supply.